Constellations: Low-Friction Exploratory Navigation with Evidence-Backed Bipartite Graphs

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Draft for discussion

Abstract

People often explore knowledge not only to answer a specific question, but to follow curiosity across many possible directions with minimal commitment: quickly trying an expansion, backing up, and trying another. We present **Constellations**, an interactive system for **low-friction exploratory navigation** that constructs **bipartite (Atomic↔Composite)** graphs on demand from natural-language queries. Users begin with a single entity and iteratively expand the graph while the system enforces bipartite alternation. To reduce instability from mid-graph ontology drift, Constellations selects a bipartite pair from the first query (currently among Person↔Event, Ingredient↔Recipe, Symptom↔Disease) and **locks it for the session** (no switching). To support interpretation and trust, Constellations attaches **evidence-backed edges** (short snippets with source links) so users can inspect the claimed reason behind a connection at any step. We describe the system design, position it relative to work on two-mode/affiliation networks and event-centric cultural heritage modeling, and propose an evaluation plan focused on recall, discovery, and sensemaking during iterative exploration. We argue that evidence-backed bipartite exploration offers a complementary paradigm to search and static knowledge graphs: a lightweight interface for navigating the frontier between what users know and what they might learn next.

1 Introduction

People often explore knowledge not to answer a single query, but to follow curiosity across many possible directions with minimal commitment: expand something, see what appears, and backtrack quickly to try a different branch. We call the boundary this process reveals the **knowledge frontier**—the set of adjacent ideas that are reachable from a current concept but not yet part of one's active memory. Interfaces for search, browsing, and encyclopedic reading support directed lookup, but can be less suited to this low-friction, branching process of "what else is nearby?" where the cost of trying a path is intentionally small.

Constellations is designed for this mode. Starting from a single user-provided entity, the system constructs a small, local graph and supports repeated **expand** operations that reveal additional connected entities. The key design decision is to enforce a **bipartite alternation constraint**: nodes alternate between **Atomic** entities (individuals or elementary concepts) and **Composite** entities (events, works, groups, conditions, or other aggregations). This constraint yields a stable interaction model across domains: atomic nodes are expanded into composites, and composites into atomics. In the project's origin domain—history—this corresponds to a simple idea closely aligned with event-centric cultural heritage modeling: **events as "meetings"** that bring multiple people together. We generalize this to other domains (e.g., films: people↔movies; cooking: ingredients↔recipes; medicine: symptoms↔diseases) without changing the core UI.

Because the system operates in open-world settings and aims for broad domain coverage, Constellations uses a large language model (LLM) to propose candidate neighbors during expansion. However, exploratory systems that rely on generative models face an immediate interpretability challenge: users naturally ask **why** a connection exists before deciding whether it is worth pursuing. To address this, Constellations attaches **edge evidence**: short evidence snippets and source links displayed when an edge is selected. Evidence is treated as support for interpretation and verification, not as proof of causal claims.

This paper contributes an interactive exploration system and a design framing for bipartite low-friction exploration. We position Constellations relative to two-mode/affiliation networks and event-centric modeling, and we outline an evaluation plan centered on recall, discovery, and sensemaking.

1.1 Contributions

- **A bipartite exploration interface** that enforces Atomic↔Composite alternation as a domain-general interaction primitive for click-to-expand exploration.
- **A session-level locking mechanism for bipartite pair selection**: the system chooses an

Atomic↔Composite pairing from the first query and locks it for the session to reduce mid-graph "ontology drift."

- **Domains + text input in one interface**: users can start from curated domain seeds or type any query, enabling both guided and open-ended exploration.
- **Evidence-backed edges** that make edge inspection a first-class interaction via short snippets and source links.
- **An LLM-assisted, on-demand graph construction pipeline** that builds local neighborhoods without requiring a precomputed knowledge graph.
- **An evaluation plan** targeting exploration outcomes (recall, discovery, orientation, and trust), rather than accuracy alone.

1.2 Why films are an especially good domain (informal motivation)

Many domains contain bipartite structure, but films provide a particularly clean and rewarding instance: people↔works with high-quality public metadata, stable naming, and strong Wikipedia coverage. This makes it easy to generate meaningful expansions (directors, co-stars, producers; films in a person's oeuvre) and to attach interpretable evidence. In contrast, domains like current politics are both more dynamic and more sensitive; they require stricter evidence policies and careful treatment of ambiguity and recency.

2 Related Work

Constellations sits at the intersection of (1) sensemaking-oriented visualization and interactive graph exploration, (2) two-mode / affiliation networks and event-centric modeling, and (3) LLM-mediated interfaces that must communicate uncertainty and evidence. We briefly summarize relevant foundations and highlight how Constellations differs in scope: it emphasizes **low-friction exploratory navigation** and **edge-level interpretability** in an open-world setting, rather than statistical inference over a fully observed two-mode matrix.

2.1 Sensemaking and exploratory visualization

Classic work on sensemaking characterizes analysis as an iterative process of foraging, externalizing structure, and revising hypotheses. This perspective motivates interfaces that make it cheap to "try a move," inspect the result, and backtrack. In InfoVis, interaction models for exploration and task taxonomies provide a language for describing what users do (e.g., search, browse, compare, derive) and for structuring evaluations beyond accuracy.

In this frame, Constellations targets tasks that are often under-emphasized in search-centric evaluation: recall, discovery, and orientation during iterative branching. The system's bipartite constraint can be viewed as a design mechanism for reducing cognitive load: it narrows the space of possible moves ("expand to the other side") while remaining expressive across domains.

2.2 Graph exploration interfaces and node-link visualization guidelines

Node-link diagrams enable intuitive local reasoning ("what connects to what?"), but scale poorly without interaction, filtering, and progressive disclosure. Design guidance for node-link diagrams emphasizes readability, avoidance of "hairballs," and interaction techniques (hover, highlight, selection, focus+context). Constellations embraces progressive disclosure by default: the graph is small and local, and growth is explicitly user-driven through expansion (including bulk expansion of frontier nodes).

2.3 Two-mode / affiliation networks (bipartite graphs)

Two-mode (bipartite) networks are a long-standing model for situations where actors participate in events or belong to groups. Canonical examples include actors↔movies, authors↔papers, boards↔companies, and other membership/affiliation structures. A key methodological lesson is that naively projecting a bipartite network into one mode (e.g., actor–actor) can introduce artifacts; bipartite structure deserves analysis and visualization in its own right.

Constellations adopts bipartiteness not primarily as an analytic device, but as an interaction constraint that simplifies exploration: each step alternates between atomic and composite nodes. Joint spatial displays of affiliation networks (e.g., correspondence-analysis-style embeddings) also provide an important precedent for "map-like" knowledge exploration and inform a future direction of large-scale knowledge cartography.

2.4 Event-centric modeling in cultural heritage and museums

Event-centric cultural heritage modeling emphasizes events as the connective tissue that relates people, places, objects, and documents. The "historical events as meetings" framing aligns strongly with Constellations' original person↔event concept: an event is any construct that brings multiple people into relation. In Constellations' current design, the system selects a bipartite Atomic↔Composite pairing from the seed term and **locks it for the session** to prevent mid-graph switching, while still enabling cross-domain use via different seeds.

2.5 LLMs, structured outputs, and evidence/provenance

LLM-based interfaces face a core tension: generative flexibility enables broad coverage, but users need interpretable reasons for what the system shows. Work on provenance and evidence in visualization motivates making "why is this here?" a first-class interaction. Constellations operationalizes this at the edge level: edges carry evidence snippets and a source link, and the UI exposes evidence on selection. This supports user judgment in a low-friction exploration loop without claiming causal conclusions.

2.6 Constellations' positioning

Constellations can be read as a pragmatic synthesis of (i) interaction design principles for progressive disclosure in node-link exploration, (ii) the representational clarity of two-mode affiliation networks, and (iii) open-world neighbor proposal via an LLM. Relative to classic two-mode network analysis, Constellations does not assume a fully observed matrix or attempt statistical inference; instead, it constructs small neighborhoods on demand and treats the bipartite constraint primarily as an interaction primitive ("expand to the other side") that limits cognitive load. Relative to static knowledge-graph browsers and search engines, Constellations emphasizes rapid branching and backtracking, and it foregrounds interpretability at the granularity where exploration decisions are made: the edge. Finally, relative to free-form LLM browsing experiences, Constellations introduces two stabilizers—bipartite alternation and a session-level lock of the Atomic↔Composite pair chosen from the seed—to reduce drift and preserve a coherent local structure during exploration.

3 System

3.1 Overview

Constellations is an interactive graph explorer that constructs a small, local graph from a user query and supports repeated expansion. The system enforces a bipartite alternation constraint:

- **Atomic nodes**: individual entities or elementary concepts (e.g., Person, Ingredient, Symptom, Player).
- **Composite nodes**: aggregations, works, events, groups, conditions, or other "meeting-like" constructs that connect multiple atomics (e.g., Movie, Recipe, Disease, Team, Historical Event/Incident).

The UI encodes this alternation with distinct visual forms (e.g., circles vs cards) while keeping interactions consistent across domains.

3.2 Data flow (high level)

1. **User query**: the user can start from a curated domain seed or type an arbitrary query.
2. **Context retrieval**: fetch lightweight context (e.g., Wikipedia summary) to help disambiguation and mitigate model knowledge gaps.
3. **Start-pair classification (locked)**: choose a bipartite pair from the first input (currently one of Person↔Event, Ingredient↔Recipe, Symptom↔Disease), then **lock it for the entire graph** (no switching).
4. **Expansion**: call the LLM to propose 8–10 neighbors on the opposite side of the bipartite partition, conditioned on the locked pair.
5. **Evidence attachment**: each proposed neighbor includes an evidence snippet + page title; selecting an edge reveals the supporting citation.
6. **Caching**: store nodes and edges (with evidence) in a database to reduce repeated calls and support persistence.

3.3 Bipartite constraint and "events as meetings"

The original domain (people↔events) is motivated by an event-centric view: an event is any construct that brings multiple people into relation. This framing generalizes naturally to other domains by choosing a Composite that aggregates multiple Atomics and for which the inverse membership relation is meaningful (e.g., actors in films; ingredients in recipes).

3.4 Evidence-backed edges

Each edge carries structured evidence:

- **kind**: currently "ai" (LLM-provided) or "none"
- **snippet**: one sentence
- **pageTitle + url**: where the snippet is claimed to come from (typically a Wikipedia page)

Selecting an edge displays this evidence in the sidebar. The goal is not to "prove" an edge, but to support interpretability and user judgment during exploration.

3.5 Figures (screenshots)

The experience is dynamic (the graph reconfigures as nodes are added; users can drag nodes to reshape local structure). A few static screenshots help convey the interaction model:

In Figure 1, expanding a composite (film) reveals connected atomic entities (people) and supporting evidence.

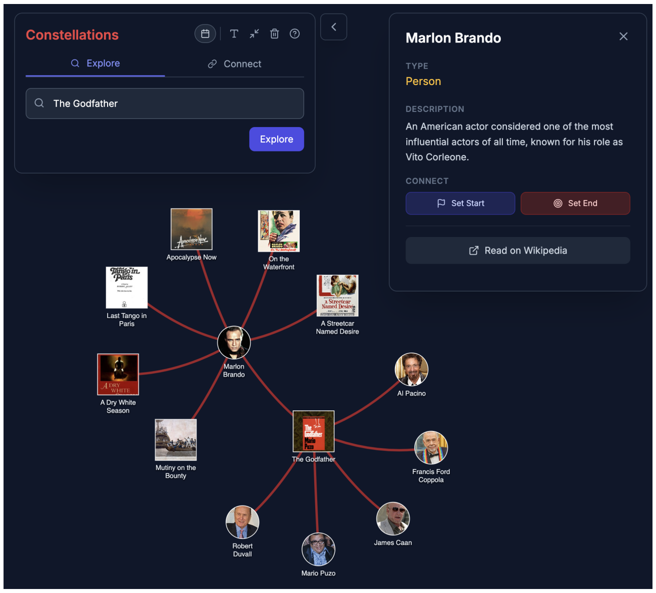

Figure 1: Figure 1. The Godfather expansion (screenshot)

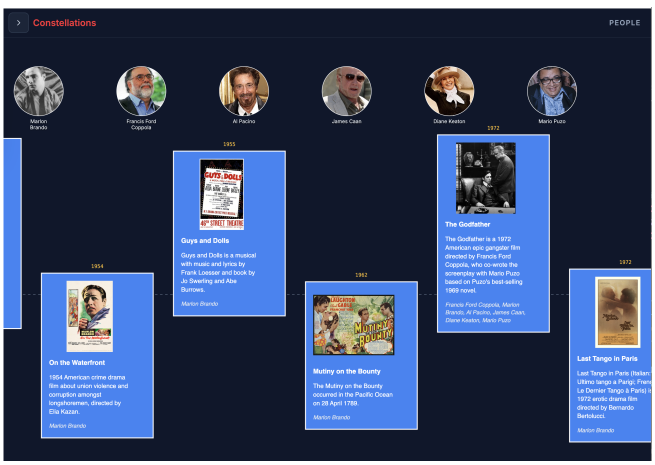

Figure 2: Figure 2. Timeline view example (screenshot)

Figure 2 illustrates the timeline view, where temporal metadata provides a second organizing lens alongside the network layout.

Figure 3 shows a path-seeking example: finding a bipartite path between two entities.

Figure 4 shows the people browser used to seed exploration.

3.5.1 Cross-domain examples (Atomic↔Composite generalization)

Figures 5–7 show the same interaction model applied across domains.

For motion, the repo includes a short demo video: `/demo.mp4`.

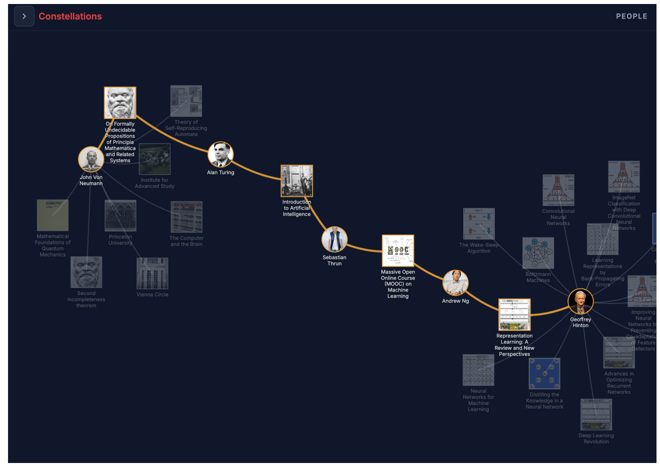

Figure 3: Figure 3. Path-seeking example (screenshot)

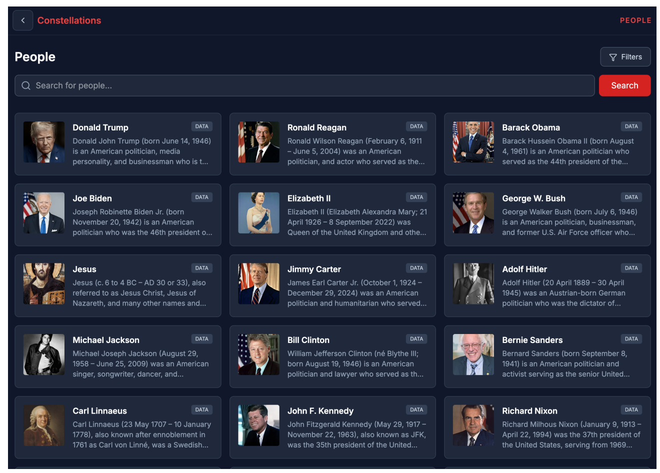

Figure 4: Figure 4. Browse People (screenshot)

3.5.2 Demo video (in-browser)

The embedded video can render as a black frame in some contexts, so we include a representative still frame instead:

Watch the demo video

3.6 Interaction design: low-friction branching

Constellations is optimized for low commitment per step:

- expanding a node is a single click,
- backing up (choosing a different node) is immediate,
- bulk operations allow quick frontier growth (e.g., expand all frontier nodes across the whole graph),
- when a node's first expansion yields very few neighbors, the UI can automatically request "expand more" once to reduce repeated clicks.

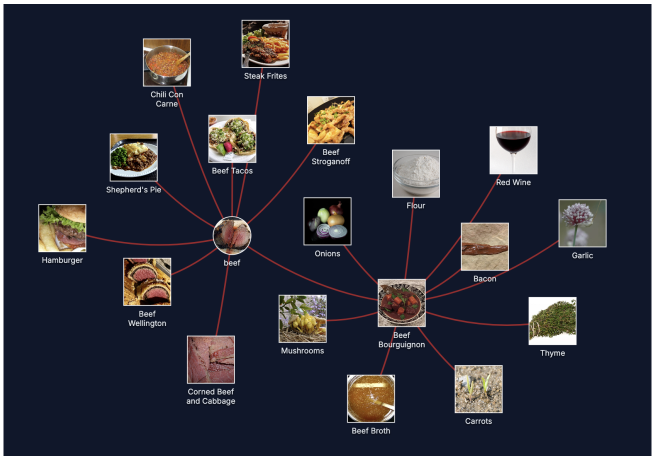

Figure 5: Figure 5. Culinary: Beef (screenshot)

Figure 6: Figure 6. Sports: LeBron James (screenshot)

This interaction model encourages "try and see" exploration, where users do not need to decide a path "once and for all."

3.7 Caching and persistence

To keep exploration responsive and reduce repeated API calls, Constellations caches:

- **nodes** (title, type, summaries, images, and classification metadata), and
- **edges** (role/label and evidence metadata).

Caching supports repeated browsing, saving/loading graphs, and revisiting a previously explored frontier with evidence intact.

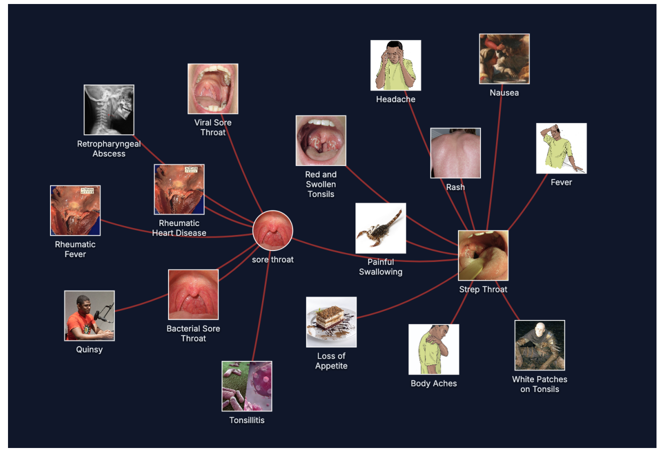

Figure 7: Figure 7. Medicine: Sore Throat (screenshot)

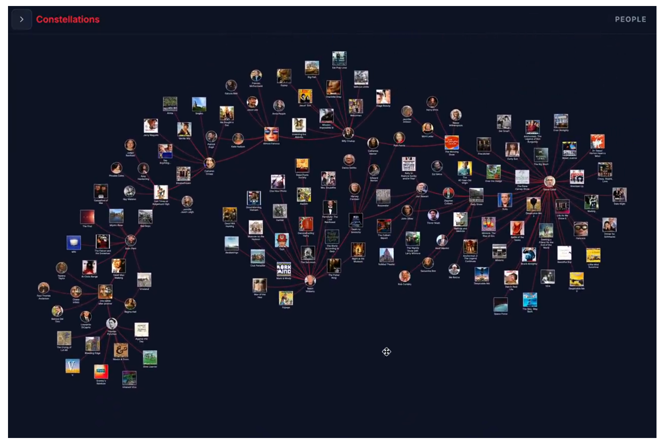

Figure 8: Figure 8. Demo frame (click through to watch the video)

3.8 Implementation notes

The current implementation is a client-side React application with a small cache backend. While the core ideas are model- and stack-agnostic, a few design choices matter for reproducibility and user experience:

- **Graph rendering and interaction**: a force-directed node-link layout supports direct manipulation (dragging nodes), hover highlighting, and edge selection for evidence inspection. Nodes are visually differentiated by partition (Atomic vs Composite) to reinforce alternation.
- **Bounded neighborhood expansion**: each expansion requests a small number of neighbors (typically 8–10) to avoid hairball growth and to keep the choice set cognitively manageable.
- **LLM output guardrails**: expansions are constrained to named entities on the opposite partition and require per-edge evidence fields (a short snippet plus a source page title/URL). This does not guarantee correctness, but it makes edges inspectable.

- **Context-first disambiguation**: before classification/expansion, the system fetches lightweight context (e.g., a Wikipedia summary) to reduce ambiguity and mitigate model knowledge gaps.
- **Session-level pair locking**: after the first query, the chosen Atomic↔Composite pair is locked for the remainder of the graph exploration to reduce mid-graph switching and preserve stable semantics for expansion prompts.
- **Caching and persistence**: nodes and edges (including evidence) are cached to reduce repeated LLM calls and to enable saving/loading graphs with evidence intact.

4 Evaluation (Plan for a First Study)

This section outlines a mixed-method evaluation aligned with **low-friction, branching exploration** rather than accuracy alone. The goal is to evaluate whether Constellations supports recall, discovery, and sensemaking in a way that is difficult to capture with conventional "answer correctness" metrics.

4.1 Study 1: Task-based user study (qual + quant)

4.1.1 Goal

Measure whether Constellations helps users:

- recall forgotten related entities,
- discover new adjacent entities,
- maintain orientation during rapid branching and backtracking,
- and assess connections using evidence.

4.1.2 Participants

- Mixed expertise: casual users + a small number of domain enthusiasts (e.g., film history or music).
- Target N=12–20 for an initial formative study; optional follow-up N=30+ for confirmatory comparisons.

4.1.3 Tasks (examples)

- **Recall**: "Start from a film/person you know. Find 5 related items you had forgotten you knew."
- **Discovery**: "Find 3 new films you want to watch next; explain why."
- **Frontier navigation**: "Keep expanding until you hit a boundary where you no longer recognize most nodes; describe what you found."
- **Evidence use**: "Given a surprising edge, use the evidence panel to decide whether you trust it enough to expand further."
- **Kiosk interaction (optional)**: "Using only taps (no typing), start from curated seeds and build a small 'story' of 8–12 nodes you would show someone else."

4.1.4 Conditions (ablation)

- With vs without **edge evidence** (hide evidence panel).
- With vs without **bipartite enforcement** (optional, if a safe relaxed mode exists) or compare to a baseline interface (e.g., search + Wikipedia navigation).
- With vs without **session-level pair locking** (when multiple pairs are enabled): measure whether locking reduces user-reported confusion and drift.

4.1.5 Measures

- Task completion (time, steps/expansions, backtracks).
- Self-reported recall/discovery (Likert + short explanations).
- Perceived orientation (NASA-TLX subset or simpler "I felt lost" scale).
- Evidence interaction frequency (edge clicks, "open source" clicks).
- Neighborhood quality proxies: duplicate rate, generic-node rate, and fraction of expansions producing ≥k meaningful neighbors.

4.1.6 Qualitative prompts (examples)

- "Describe a moment where you changed direction quickly—what triggered it?"
- "Describe a moment where an edge's evidence changed your decision to explore further."
- "What did the system surface that you felt was 'on the edge' of your memory?"
- "Did you notice the Atomic↔Composite alternation? Did it help you predict what would happen when you clicked?"

4.2 Study 2: Log-based / offline quality evaluation

4.2.1 Goal

Assess properties of generated neighborhoods without claiming "truth":

- **Named-entity quality** (avoid generic phrases).

- **Diversity** (avoid near-duplicates).

- **Evidence availability** (fraction of edges with evidence snippets).

- **Stability under repeated use** (cache hits; consistency of expansions for the same seed).

4.2.2 Sample design

Select a set of seeds across domains (films prioritized), run multiple expansions, and evaluate with:

- rater checks (human annotation),

- plus simple structural metrics (degree distributions; repetition).

4.3 Reporting "knowledge frontier"

One paper-specific contribution could be operationalizing the "frontier":

- Track the point during exploration where a participant reports low recognition.

- Quantify as a function of depth/expansions and node familiarity labels.

4.4 Analysis plan (lightweight)

- Use within-subject comparisons where possible (evidence on/off; lock on/off), counterbalanced across participants.

- Quantitative: paired tests on time/steps/backtracks and Likert outcomes; report effect sizes with confidence intervals.

- Qualitative: thematic analysis of "frontier" moments, evidence-driven decisions, and breakdowns (ambiguity, drift, low-signal expansions).

5 Discussion & Future Work

5.1 Discussion: what Constellations is (and is not)

Constellations is designed for **exploration**, not inference. While bipartite network analysis offers statistical models for affiliation data, Constellations operates in an open-world setting with locally constructed neighborhoods and focuses on user experience: recall, curiosity, and sensemaking.

5.2 Limitations

- **Open-world ambiguity and name collisions**: many entities share names (e.g., people vs works with the same title). Lightweight context helps, but disambiguation remains imperfect and can propagate errors into expansion.

- **Model knowledge gaps and recency**: LLMs can be outdated, and Wikipedia coverage varies; the system can misclassify entities or miss important neighbors without stronger retrieval and verification.

- **Evidence is supportive, not definitive**: a single snippet and link is often enough for user judgment, but it does not guarantee the edge is correct. Automated snippet verification and multi-source provenance are not yet implemented.

- **Domain variance**: some domains naturally form clean affiliation structures (e.g., actors↔films), while others exhibit weaker "membership" semantics or higher noise. Session-level pair locking reduces drift but can also constrain legitimate cross-type exploration.

- **Scalability and layout stability**: force-directed layouts are effective for small local neighborhoods but can become unstable or visually dense under repeated bulk expansion, motivating multi-resolution views and clustering.

5.3 Future work directions

5.3.1 Better expansion ranking (bipartite-aware)

Use bipartite-inspired heuristics for ranking and diversity:

- down-weight generic hubs,

- promote diversity across roles/decades/types,

- rank by evidence quality and specificity.

5.3.2 Stronger evidence and provenance

Move from "one snippet" to richer provenance:

- multiple evidence items per edge,

- automated verification that the snippet exists in the claimed source,

- user feedback loops (confirm/reject edges).

5.3.3 Pair learning and "soft locks"

The current system chooses among a small set of bipartite pairs and locks the choice for the session. A natural extension is to treat pair selection as a probabilistic belief state: maintain a small set of candidate pairings with confidence, allow controlled switching only when confidence crosses a threshold, and preserve previously drawn edges by freezing node partitions once placed.

5.3.4 Global "map-like" exploration (joint embedding)

Inspired by joint displays of affiliation networks, a long-term direction is to place a very large two-mode graph into a shared 2D space ("knowledge cartography"), enabling map-like exploration at multiple zoom levels. A practical approach would require multi-resolution embeddings and approximate optimization rather than exact global minimization.

5.3.5 Domain packs and curatorial modes

For domains like film and museums:

- curated constraints on allowed composite types,
- exhibit-specific "packs" with guided seeds and narratives,
- touch-screen / installation mode for public spaces.

5.3.6 Scaling limits as a design probe

Bulk frontier expansion (e.g., expanding all leaf nodes repeatedly) provides a practical way to probe system limits: at what point do layout stability, latency, and evidence quality degrade? Characterizing this "interesting limit" could inform adaptive strategies such as multi-resolution views, clustering, and progressive summarization.

References (Working List)

This is a lightweight, link-forward reference list for the "printable snapshot." For the evolving bibliography, see `../bibliography.html`.

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